CASE STUDY PRESENTATION

Satellite Telemetry Anomaly Detection Using Machine Learning in R

Case Study Presentation

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Background

Satellite Telemetry

Generated by spacecraft sensors and subsystems to report the health and behavior of the satellite.

Abnormal Behavior

Telemetry segments may contain unusual or abnormal signal behavior that needs attention.

Manual Monitoring

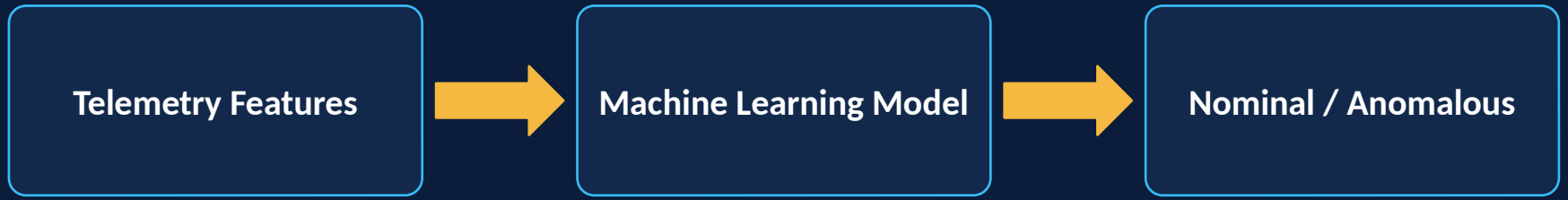
Reviewing large volumes of telemetry by hand is time-consuming.

Machine Learning

ML can help classify telemetry segments as nominal or anomalous.

Problem Statement

“To identify anomalous satellite telemetry segments from nominal telemetry using machine learning in R.”



Task: telemetry-segment anomaly classification (not future failure forecasting).

Objectives

1. To understand and analyze the official OPSSAT-AD satellite telemetry dataset.
2. To clean and preprocess the telemetry features using R.
3. To develop machine-learning models for detecting anomalous telemetry segments.
4. To evaluate and visualize model performance using suitable classification metrics.

Dataset: OPSSAT-AD

2,123	23	9	18
Rows (segments)	Columns	Channels	Handcrafted features

1,594	529	1,689	434
Training	Testing	Nominal	Anomalous

Source & Target

OPSSAT-AD – anomaly detection dataset for satellite telemetry, Zenodo Record 12588359, Version 1.

DOI: 10.5281/zenodo.12588359

Target: anomaly (0 = Nominal, 1 = Anomalous)

Why It Is Suitable

- Real telemetry from ESA's OPS-SAT CubeSat
- Ground-truth anomaly labels
- Official train/test split
- Telemetry-derived features for ML
- Official v1 file checksum verified in notebook

Important Variables

Metadata

- segment
- train
- channel
- sampling

Target

- anomaly
- 0 = Nominal
- 1 = Anomalous

Telemetry Features (18)

- duration, len
- mean, var, std
- kurtosis, skew
- n_peaks
- Smoothed peak-count features
- Difference-based peak features
- Difference-variance features
- Gap-related features
- Length-derived features
- Variance-derived features

Data Preparation in R

Workflow

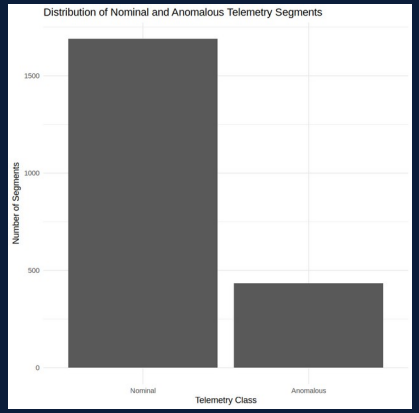
1. Load data
2. Inspect structure
3. Missing-value check
4. Duplicate check
5. Non-finite-value check
6. Factor conversion
7. Official train/test split
8. Training-only median imputation (if required)
9. Remove identifier/split columns from predictors

```
data$anomaly <- factor(data$anomaly,
  levels = c(0, 1),
  labels = c("Nominal", "Anomalous"))
data$channel <- factor(data$channel)
train_data <- data[data$train == 1, ]
test_data  <- data[data$train == 0, ]
predictor_cols <- setdiff(names(data),
  c("segment", "anomaly", "train"))
```

Key Rules & Check Results

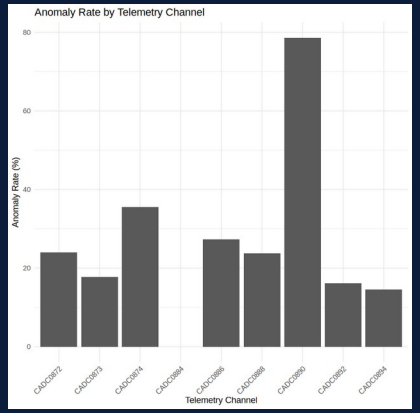
- segment and train excluded from predictors
- anomaly → factor; channel → categorical
- Checks: 0 missing, 0 duplicate, 0 non-finite
- Training-only median imputation: nothing to fill
- Train 1,594 (321 anomalous); Test 529 (113 anomalous)

Data Analysis & Visualization



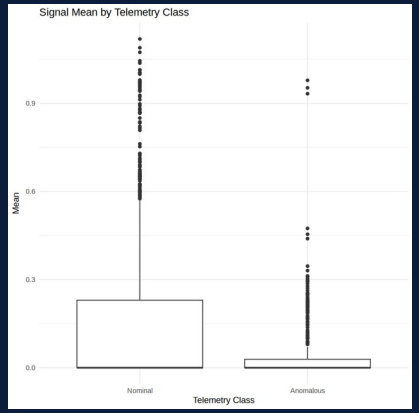
Class distribution

Nominal 1,689 vs anomalous 434: about 4:1, so the classes are imbalanced and accuracy alone is not enough.



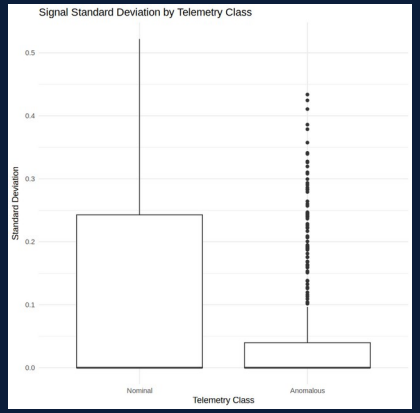
Anomaly rate by channel

Rates differ by channel: CADC0890 is highest (78.6%, only 14 segments); CADC0884 has none; CADC0874 is 35.6%.



Signal mean by class

Both medians are near zero. Nominal means spread wider (75th percentile ≈ 0.23 vs ≈ 0.03 for anomalous).



Signal std by class

Same pattern: nominal std spreads wider (75th percentile ≈ 0.24 vs ≈ 0.04); both medians are near zero.

Machine Learning Method

Primary: Random Forest

- Captures nonlinear relationships
- Combines many decision trees
- Suited to classification
- Robust across many telemetry features
- Provides feature importance
- 500 trees, 4 variables tried per split

Comparison: Logistic Regression

- Interpretable statistical baseline
- Useful for model comparison
- Simpler linear classification model

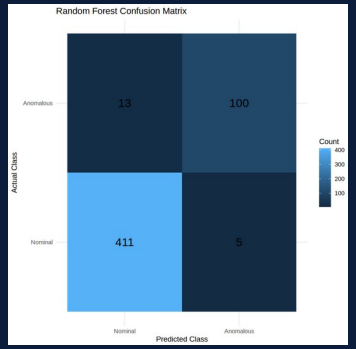
Supervised binary classification: OPSSAT-AD provides ground-truth anomaly labels.

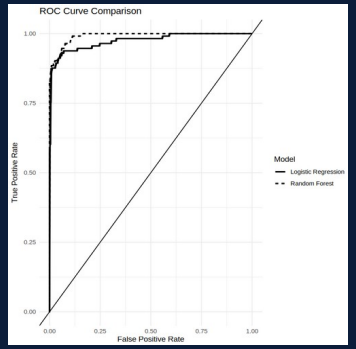
Model Evaluation

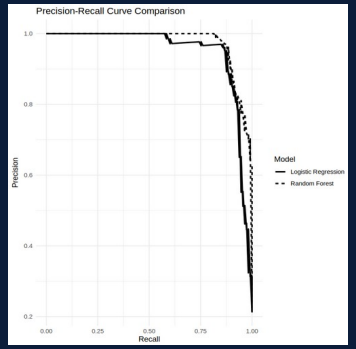
Metric	Random Forest	Logistic Regression
Accuracy	**0.9660**	0.9641
Precision	0.9524	**0.9700**
Recall	**0.8850**	0.8584
F1-score	**0.9174**	0.9108
MCC	**0.8970**	0.8910
ROC-AUC	**0.9914**	0.9743
PR-AUC	**0.9747**	0.9489

Selected: Random Forest

Best F1 and PR-AUC in the notebook. Green = better model per metric.



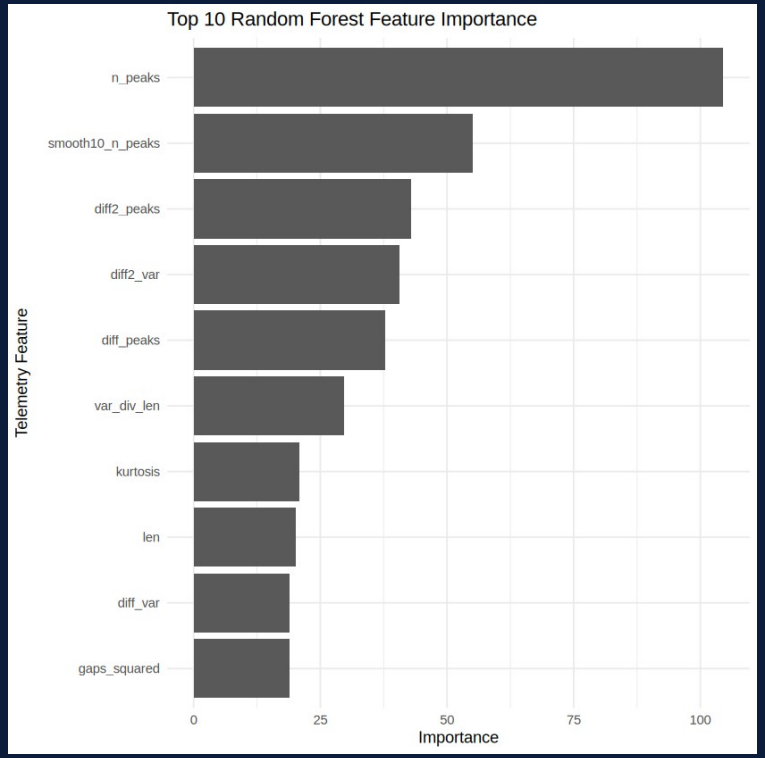




Reading the Results (test set, 529 segments)

- RF confusion matrix: 411 true nominal, 5 false alarms, 13 missed anomalies, 100 detected anomalies
- RF is higher on 6 of 7 metrics; Logistic Regression has higher precision (0.9700 vs 0.9524)
- ROC-AUC: 0.9914 (RF) vs 0.9743 (LR); PR-AUC: 0.9747 vs 0.9489
- MCC of 0.8970 shows strong agreement despite class imbalance

Feature Importance & Findings



What the Random Forest Uses

- n_peaks ranks first (importance 104.5), almost twice the next feature, smooth10_n_peaks (55.1)
- Peak-count features lead; diff2_var ranks 4th and diff_peaks 5th
- mean and std are not in the top 10
- Importance = mean decrease in Gini (500 trees)
- False positives: 5 nominal segments flagged as anomalous (false alarms)
- False negatives: 13 anomalous segments missed
- More misses than false alarms, matching recall 0.8850 < precision 0.9524

Conclusion, Reproducibility, Version Control & References

Conclusion

- Official OPSSAT-AD dataset analyzed
- R-based cleaning and preprocessing performed
- EDA and visualization completed
- Random Forest and Logistic Regression evaluated
- Imbalance-aware metrics used; Random Forest best (F1 0.9174, PR-AUC 0.9747)
- Result: a telemetry-segment classifier, not future-failure forecasting

R Markdown & Reproducibility

- Code, outputs, tables, graphs organized in R Markdown
- Run in Google Colab (R runtime); rerun to reproduce

Version Control

- v1_initial_dataset_and_problem
- v2_preprocessing_and_eda
- v3_model_development • v4_final_results

References

1. Ruszczak et al. (2025). The OPS-SAT benchmark for detecting anomalies in satellite telemetry. Scientific Data. DOI: 10.1038/s41597-025-05035-3
2. OPSSAT-AD, Zenodo Record 12588359, v1. DOI: 10.5281/zenodo.12588359
3. KP Labs OPS-SAT-AD GitHub repository.